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ORIGINAL RESEARCH

Predicting Financial Distress Using Machine Learning Techniques

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Abstract

Bankruptcy in future would lead to heavy losses, and attempts should be made to reduce it and prevent such a loss in advance. Potential misclassification of potential and futurist bankruptcy can be referred to as an audit failure. Predicting bankruptcy for different users, including investors, auditors, creditors and regulators, is essential. Various prediction models have been considered in other studies, and this research study includes multiple techniques like random forest, ANN, and logistic regression. The main reason behind the prediction of financial distress is that it will help ensure an increase in compatibility with the decision-making process. In this study, 1111 companies were considered liquidated and restructured by NCLT since its inception, i.e., 2016. Of these, 342 companies had their resolution plan approved from 2017 to 18 till December 2023, while 769 companies were liquidated. The selected companies' financial information has been considered for the last five years, and machine learning (Random forest, Artificial neural network) and traditional models (Logistic regression) have been used to predict the outcome of the firm (in terms of bankruptcy or restructuring) with their respective level of accuracy. The study, firstly, reveals that machine learning models have better predictive accuracy than statistical models. Secondly, the model's predictive accuracy is highest near the year/event of distress; as we move further from the year of liquidation/restructuring, the accuracy declines. Thirdly, financial variables and firm-specific characteristics have better precision and accuracy than only assessing the firms based on economic parameters.

Keywords Financial distress · Liquidation · Machine learning · Prediction · Restructuring

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1 Introduction

Bankruptcy exerts a detrimental impact on any corporation. Various methodologies can predict bankruptcy, which is crucial for diverse users, including investors, auditors, creditors, and regulators. Future bankruptcy could result in substantial losses, necessitating proactive measures to mitigate and preempt such losses. Misclassification of potential and future bankruptcy could lead to allocating scarce financial resources to inefficient firms, further leading to the erosion of public finance. Consequently, there is a need to enhance the accuracy of bankruptcy prediction. Techniques need to be developed to ensure stakeholders derive certain benefits.

Business failure is not an abrupt outcome or event; it is a process where firms undergo various stages of financial distress before liquidation or being declared bankrupt. This underscores the importance of early detection signs, enabling management to take corrective actions and avert failure. Intuitive credit risk management benefits all stakeholders: lenders, management, shareholders, investors, and regulators. Basel III norms re-emphasise the importance of credit risk management by developing credit risk models, which would, in turn, lead to decreased delinquencies in bank loans and Non-performing assets (NPA). Early warning models are widely used to evaluate a company's financial health; however, this analysis should be contextualized within value creation, which is the primary objective of firms operating in capital markets. Conservative corporate strategies are often viewed as safer, lowering both risk and returns, whereas aggressive strategies are associated with higher levels of risk and returns, as indicated by models like the Altman Z-Score (Bolek & Gniadkowska-Szymańska, 2023).

While there is existing literature on predicting financial distress, there remains a need to reevaluate the models used for such predictions. Credit risk models, which have been used interchangeably for insolvency and bankruptcy prediction and financial distress prediction (Gepp et al., 2010), initially employed Logistic regression (LR) for distress prediction. However, with Artificial Intelligence techniques such as Artificial neural networks and Support vector machines, predictive modelling has become a crucial tool for decision-making (Abu-Mostafa, 2001).

This research contributes to developing financial distress prediction models by utilising parsimonious information, including financial ratios derived from the economic data provided by firms. The quantitative variables considered include leverage, liquidity, and corporate profitability. Various prediction models have been explored in different studies; this research incorporates techniques like Decision trees, Random forests, and Artificial neural networks (ANN) and conducts a comparative analysis with the traditional method of Logistic Regression. The study has been performed on 1111 firms that were under financial distress and were liquidated or underwent restructuring through NCLT from 2017 to 2023.

The trade-off theory posits that firms must balance the costs and benefits associated with debt usage. Key advantages of debt financing include the interest tax shield, reduced financing costs, and preservation of ownership structure, while the associated costs encompass bankruptcy expenses and the likelihood of

financial distress. This necessitates robust credit risk management by lenders and financial institutions (Suresh et al., 2023). Credit risk management is of utmost concern in India, also due to the alarming rate at which the NPAs have risen over the last several years. This led the Central bank, i.e. The Reserve Bank of India, to develop various schemes to reduce the exposure of commercial banks to ever-escalating issues of NPAs. Corporate debt defaults were governed by different laws and regulations before 2016, like the Sick Industrial Companies Act (SICA), 1985, Debt Recovery Act, 1993, Corporate Debt Restructuring (CDR), Strategic Debt Restructuring (SDR), Securitization and Reconstruction of Financial Assets and Enforcement of Security Interest Act (SARFAESI), 2002 and Company Law, 2013 (amendments to Companies Act). These laws and regulations failed to solve the banks' twin balance-sheet problem, which led to the conversion of standard assets (loans) of the banks to sub-standard assets and finally into NPAs. The continuing increase in the number of NPAs has advocated the implementation of a uniform code to recover and resolve stressed assets. Insolvency and Bankruptcy Code (IBC) was implemented in India in 2016. As per IBC 2016, companies undergoing financial distress can approach the National Company Law Tribunal (NCLT), the legal body that would appoint insolvency professionals (IP) to manage the process. The result would be either restructuring the stressed assets or liquidating the company under trial. (Gupta, 2022).

This study evaluates the predictive ability of Logistic Regression and two machine learning models (Random Forest and Artificial neural network). The purpose is to find a suitable model for predicting whether a company can be restructured or needs to be liquidated so that the misclassification can be eliminated. The main reason behind this is that stakeholders need to decide while classifying firms that are financially distressed and are potential candidates that could face bankruptcy in future. This would ensure an increase in compatibility towards the decision-making process.

The analysis indicates that predictive accuracy is enhanced when firm-specific characteristics, such as age, industry, and size, are incorporated alongside financial variables. Among the models evaluated, Random Forest demonstrated the highest efficacy in predicting financial distress, followed by neural networks and logistic regression, suggesting that machine learning models generally surpass traditional statistical models for this application. Additionally, company size was identified as a significant determinant of distress outcomes, with smaller firms being more susceptible to liquidation, a finding consistent with prior studies by Almaskati et al., (2021) and Sehgal et al., (2021). Furthermore, the predictive accuracy of all models declined as the forecasting horizon increased, a trend similarly reported by Charalambakis and Garrett (2019) and Sehgal et al., (2021). Key financial variables contributing to distress prediction include Profit After Tax as a percentage of total income, Return on Assets, and Interest Coverage.

The rest of the paper proceeds as follows: Sect. 2 reviews the literature, Sect. 3 describes the study's objective, Sect. 4 highlights the data and methodology, and Sect. 5 provides the results and discussion. Finally, Sect. 6 concludes and provides the scope for future research.

2 Review of Literature

2.1 Studies Employing Statistical Model for Predicting Distress

Since the 1960s, many types of research have focused on companies that have become bankrupt. "Insolvency" refers to a company's financial situation when it finds it difficult to generate successful performance to satisfy the liabilities. The companies conduct financial analyses to ensure long-term viability. The study warns about potential risk and solvency issues by analysing a company's financial performance through its liquidity, financial stability and creditability. Many studies have been conducted in different fields to adapt the insolvency prediction model for businesses in other sectors. Research has been conducted to predict the solvency of various firms. The previous researchers incorporate both financial and non-financial aspects in computation. Studies mainly focus on financial ratios derived from financial reports instead of non-financial elements like a firm's age or management expertise.

In many published studies, financial ratios have been used to forecast solvency. Beaver (1966) analysed thirty ratios split into six groups. The ratio was chosen from different groups, which specified various solvency problems like current ratio, total debt/ cash flow, etc. Altman (1968) used the Z-score model to predict financial bankruptcy. The study used different variables estimated using a balance sheet and income statement. Z-score models indicated the probability of avoiding failure. The variables considered for evaluating the firms' financial conditions were coverage, activity, leverage, profitability and liquidity (Altman et al., 2010). Zmijewski (1984) developed a weighted probit model for predicting bankruptcy. Hopwood et al. (1994) noted that financial distress is related to a probability of going bankrupt. Rather than using a multiplicity of various financial ratios used in previous research, this study is based on specific models, including logistic regression, ANN, etc. Some earlier studies (DeFond and Jiambalvo, 1994) have considered a model incorporating various ratios that helped measure liquidity, solvency, and profitability. The author selected the model based on a quantitative measure named the Zmijewski score.

Various research studies have indicated the use of qualitative measures of financial distress. As per evidence provided in research studies by Gajpal et al., (1994), Gilson et al. (1990), Giroux and Wiggins (1984) and Turetsky (1997), there is use of negative operating cash flow, omitting or reducing dividend payment, debt default includes default in loan payments or technical default. Troubled debt restructuring has also been used to indicate financial distress before a firm goes bankrupt. The study has used qualitative variables as dummy variables.

John and John (1993) suggested that financial distress could progressively result in negative operating cash flows. Giroux and Wiggins (1984) consider negative cash flows signalling a decline in corporate financial status. In the study, Foster et al., (1998) mentions that any firm's cash flow provides vital information to the investors and helps evaluate the chances of economic distress. As per the author's opinion, negative operating cash flow implies a tendency towards

insufficient working capital and future cash position. This could lead to the survival of the firm in the long term.

Dielman and Oppenheimer (1984) noted that a firm's dividend decisions could be vital in identifying financially distressed firms. The authors mentioned in the study that firms generally follow a stable dividend policy when the firm is financially healthy. DeAngelo et al., (1994) contended that any firm facing financial problems could think of omitting or cutting dividends. Hence, dividend changes, mainly reductions in the amount of dividends, might predict financial distress. Dividend omissions or reductions were measured as dummy variables.

Chen & Church (1996) have mentioned that when a firm is about to fail, its financial ratios deteriorate. However, a clear indication of potential problems could lead to bankruptcy, and it becomes challenging to encounter debt obligations. It is not easy to comply with lending agreements and make scheduled payments. Additionally, when certain firms violate the debt covenants, the creditors might accelerate debt maturity and impose higher re-contracting costs on the lenders (Anandarajan et al., 2001). Further, any technical defaults would constrain the freedom of action while operating in the firm (Smith & Warner, 1979). When the firm defaults on the debt agreements and cannot provide a remedy through re-negotiation or doesn't get a waiver from the creditors, there is a higher chance of becoming bankrupt when the company defaults (Varadraj & Abhay, 2014).

Previous studies have regarded Troubled debt restructurings (TDR) as a distinct stage in the financial distress continuum (as mentioned by Giroux & Wiggins, 1983 and Turetsky, 1997). TDR results from various private debt renegotiations to resolve or remedy debt defaults (Anandarajan et al., 2001). As a result of TDR, current and future cash flows related to debt service might improve. TDR firms that are unsuccessful might end their bankruptcy. TDR could be considered a clear signal about financial distress, which could indicate bankruptcy.

As per Ooghe and Prijcker, (2008), lower financial independence, weaker profitability, increased expenses and lower cash flows were considered the critical solvency predictors. The study observed increased expenditure at constant turnover, which might help predict financial problems and inadequate professional competence for implementing decisions arising from various non–financial predictors. Bhimani et al. (2010) applied eleven financial ratios: financial coverage, capital/ total asset, investment ratio, days in payables and receivables, interest cost, financial coverage, solidity, gross income and return on equity. They have highlighted days and payables in receivables that determine the payment behaviour of creditors and debtors. In a study of Taiwan electronics market by Su and Huang (2010), firm failure prediction for Taiwan's electronics companies is conducted using the Logit regression model, the KMV option-based asset valuation model, and the ZPP GARCH autoregressive time series simulation model. It was concluded that the predicted failure probabilities are much larger using the Logit model than the other models.

Al-Kassar and Soileau (2014) selected specific ratios based on assets and liabilities, and there was no credit interval for failure in prediction. The study indicated that the higher the profitability and other ratios, the lower the solvency risk. When this ratio is lower, the risk of solvency is high. Mironiuc et al., (2015) indicated that the current ratio, average collection, financial leverage, liquidity, financial expenses,

and other ratios indicated that factors were significant statistically in predicting the firm's solvency position. Liang et al., (2016) have used ratios like cash flow, growth, turnover, profitability, solvency, and capital structure to assess bankruptcy. The study observed profitable and solvency ratios as the most significant ratios to predict solvency.

2.2 Studies Employing Machine Learning to Predict Distress

In the current times, machine learning (ML) is considered to be a cross between Computer Science and Statistics. Artificial INTELLIGENCE (AI) has been constantly improved, and big data is easily accessible in different sectors. The computation cost has been reduced, and people can use AI in the best way possible. AI's evolution directly impacts the technology market, mainly the firms related to AI (Lu et al., 2018).

Network structures with more robust mapping capacity could manage the interaction between non-linear associations and variables. Artificial neural networks (ANNs) are a solution for different Machine learning (ML) models. They are used in the financial sector to predict exchange rates, financial distress, credit scoring, and portfolio management.

Among the early studies, Henley and Hand (1996) used the k-nearest neighbour (k-NN) algorithm to assess the customer's ability to pay loans. Galindo and Tamayo (2000) performed a comparative examination of machine learning and statistics to determine mortgage loan defaults. They concluded that the best estimates were provided by CART Decision trees, followed by neural networks and k-NN.

Anandarajan et al., (2001) used an artificial neural network to predict the bankruptcy of financially distressed firms. They used the Zmijewski distress score, which incorporates ratios measuring profitability, liquidity, and solvency.

Yeh and Lien (2009) examined the forecasting accuracy of a few data mining techniques and observed that artificial neural networks (ANN) estimated the default more accurately than the other models.

Decision trees were used by Gepp et al., (2010) to predict business failure, and they observed that decision trees were far superior to traditional discriminant analysis. Similarly, Bhandari and Iyer (2013) used ratios on 50 failed and 50 non-failed US firms. They concluded that discriminant analysis was a better predictor than the cross-validation approach adopted under the data mining technique. Li et al. (2014) evaluated distress and the effectiveness of financial ratios using data envelopment analysis and logistic regression on a dataset of Chinese enterprises. Their results confirmed the theory that using efficiency information improves prediction ability.

Morum and Roy (2013) employed seven financial ratios to predict corporate sickness in the Indian Steel industry. They developed the model using stepwise logistic regression and cluster analysis on 40 companies classified as sick and non-sick. The authors believed that stepwise logistic regression has a better predictive ability than the other methods.

In the study, Bapat & Nagale (2014) compared the predictive powers of three distinct distress models for Indian listed companies: discriminant, logistic, and neural

network. They opined that neural networks were the most accurate regarding precise classification.

Behr and Weinblat (2017) conducted another study using machine learning tools, logit regression, decision trees, and random forests. The study included German, French, Italian, British, Spanish, and Portuguese firms. They observed that distressed firms had a spiralling debt ratio, liquidity ratio, return on sales and return on assets, which were significant bankruptcy predictors.

Khemakhem and Boujelbene (2018) integrated financial and non-financial data to create a credit score model. Their research revealed that profitability ratios, repayment capability, and solvency are essential default indicators.

Addo et al., (2018) worked on binary classifiers using machine learning models to forecast loan default probabilities. They concluded that decision trees are more accurate than neural networks for predicting distress.

Shrivastava et al. (2018) predicted a corporate crisis for enterprises in India that did not have bankruptcy specifics. Based on their data, they argued that the Bayesian technique performed better in accuracy and predictive ability than the standard logistic model. In another study in the same year by Nair and Sachdeva (2016) on 574 Indian manufacturing companies classified as distressed and non-distressed, it was established that profitability and efficiency ratios were of utmost importance while predicting distress. Tang et al., (2019) used the random forest model to ascertain the credit card default risk for the firms involved in China's energy business. Their results indicated that the overdraft ratio and the amount of credit card costs significantly affect credit risk. De Paula et al., (2020) analysed data from a Brazilian credit union's loan transactions for credit risk. They established the superiority of the random forest technique in estimating credit scoring compared to logit regression. Shrivastava et al., (2020) employed logistic regression, random forest, and Ada Boost to predict bank failure. Ada Boost provided the highest level of accuracy when compared to the other approaches.

Golbayani et al. (2020) used four machine learning techniques deemed helpful in previous studies (bagged decision trees, random forest, support vector machine, and multilayer perceptron) on a dataset, and the results show that decision tree-based models perform better on the chosen datasets.

Recently, a study of 12,000 small and Medium enterprises concluded that the Random Forest algorithm had the highest prediction accuracy compared to Logistic regression and Artificial Neural networks (Malakauskas & Lakštutienė, 2021).

Gupta (2022) evaluated four machine learning models on the companies under IBC and revealed that random forest outperforms the other three techniques: Support Vector Machine, Decision tree and Logit model.

A study was performed to predict financial distress amongst Fintech companies by employing an artificial neural network model. The study pinpointed Return on capital, current ratio, quick ratio and debt-to-equity ratio as significant predictors of financial distress. (Halteh et al., 2024).

Alvi and Arif, (2024) undertook a thorough longitudinal analysis of non-financial firms in Pakistan and an extensive assessment of machine learning algorithms for predicting default risk. The lack of early warning indicators within the non-financial

sector contributes to insolvency and financial distress, which in turn adversely affects the economy and drives higher unemployment rates.

Most studies employing data mining techniques opine that machine learning models outperform statistical models in terms of discriminatory power and precision. It was also discovered that machine learning algorithms for credit distribution led to allocation to safer and larger borrowers, reducing lenders' losses.

The current study intends to bridge the gap in the existing literature by comparing the predictive accuracy of statistical models and machine learning models on companies under IBC to predict bankruptcy or reorganisation. It is imperative to develop specific models based on assessing the likelihood of defaulting on a counterparty. The financial institutions deal with counterparty default now and then, a tragic issue they face. The lenders, financial market players, and government should reduce counterparty defaults to protect the creditor's interest and restore a firm's sustainability. Financial Distress Prediction intends to solve problems by classification through financial ratios, which could describe a firm's current situation for a particular period. Also, there is a need to identify the economic and firm-specific factors that can serve as an early indicator of default.

3 Objectives of the Study

The current study caters to the following objectives-

- To predict whether the firm will become bankrupt or be restructured by NCLT based on the analysis of financial information in the form of financial ratios using Machine Learning Techniques and Statistical Techniques.
- To understand the key ratios that define the impending default
- To identify the impact of non-financial variables like age, size of the firm and industry group on the predicting default.
- To assess the predictive ability of model(s) years before the restructuring/liquidation.

4 Data and Research Methodology

This study is analytical in nature. It focused on analysing five years of data from 1111 companies.

Most published studies have used healthy firms as the control group. In this study, 769 companies were considered liquidated by NCLT since its inception, i.e., 2016, and 342 companies had their resolution plan approved from 2017 to 18 till December 2023. Table 1 shows the details of the companies included in the study's sample. The selected companies' financial information has been considered for five years before being liquidated or restructured. We used financially stressed firms admitted to NCLT as the sample of study to ensure increased compatibility to a realistic decision-making process as adopted by Anandarajan et al., 2001. This is because

Table 1 Year-wise data of restructured and liquidated companies (source-IBBI)

Year	No. of cases admitted in IBBI for restructuring	No. of restructured cases available in CMIE prowess	No. of cases admitted in IBBI for liquidation	No. of liquidated cases available in CMIE prowess	Total Number of distressed firms for which the data is available	Financial and Non-Financial data collected for the years
2016–17	Nil	**Nil**	23	**16**	16	2011–12 to 2015–16
2017–18	19	**7**	376	**158**	165	2012–13 to 2016–17
2018–19	75	**31**	550	**195**	226	2013–14 to 2017–18
2019–20	132	**46**	751	**214**	260	2014–15 to 2018–19
2020–21	119	**42**	161	**43**	85	2015–16 to 2019–20
2021–22	144	**46**	231	**58**	104	2016–17 to 2020–21
2022–23	189	**72**	155	**20**	92	2017–18 to 2021–22
2023–24 (Till December 2023)	208	**98**	354	**65**	163	2018–19 to 2022–23

The bold text signifies the best out of all the values signifying the superiority of a model over the others based on that specific parameter

the Resolution Professionals have to decide to classify a financially distressed firm (rather than a financially healthy firm) as a potential candidate for bankruptcy.

The observations regarding 1111 companies for the past five years gave us 5555 firm observations, which were used to identify the key financial ratios that predict the outcome of economic distress.

Table 2 gives a snapshot of sample characteristics from various industries.

The industry classification is based on the National Industrial Classification (NIC) 2008, adopted by the Government of India to disseminate industry-wise economic data.

Table 3 Based on an extensive literature review, the financial ratios and the independent variables employed in the study.

The literature review shows that machine learning has been used to predict bankruptcy and credit risk management since the 1990s. Statistical tools, which relied more on assumptions of normality and linearity of data, gave way to machine learning techniques, which were data-driven, robust, and distribution-free (Gupta, 2022).

Statistical Techniques were predominant from the 1960s until the 1990s, with Multivariate discriminant analysis (MDA), proposed by Altman (1968), and Logit analysis (LA) being the prominent ones (Bellovary et al., 2007). Altman used six financial ratios from the companies' financial statements to measure risk, and since then, numerous studies have used the Altman MDA model. (Martin, 1977) Developed an early warning model predicting bank failure using Logit Analysis and compared it with Linear Discriminant Analysis.

In addition to these, Ohlson (1980), Jones and Hensher (2004), and Canbas et al., (2005) used more improved and innovative modelling techniques. Statistical

Table 2 Various industries to which the sample companies belong

S. no	Industry	No. of companies restructured	No. of companies liquidated	Total
1	Mining and Quarrying	01	Nil	01
2	Manufacturing	157	361	518
3	Electricity, Gas, Steam and Air conditioning Supply	15	24	39
4	Construction	37	62	99
5	Wholesale and Retail Trade: Repair of Motor Vehicles and Motorcycles	44	164	208
6	Transportation and Storage	10	26	36
7	Accommodation and Food Services Act	8	5	13
8	Information and Communication	16	18	34
9	Financial and Insurance Activities	36	67	103
10	Professional, Scientific and Technical Activity	1	12	13
11	Administrative and Support Service Activities	13	18	31
12	Education	1	1	2
13	Human Health and Social Work Activities	3	6	9
14	Arts, Entertainment and Recreation	0	3	3
15	Other Service Activities	0	2	2

Table 3 Financial variables and firm-specific characteristics included in the study

Financial variables	
Measure	Ratio/Metric
Efficiency	Total asset utilisation ratio Finished goods turnover ratio Debtors Turnover ratio Creditors Turnover Ratio Gross Fixed Asset Utilisation Ratio Return on Total Assets Return on Capital Employed Return on Net Worth
Profitability	Profit before tax as % of total Income Profit After tax as % of total Income Net profit margin Operating Profit margin Cash profit as % of Total Income
Coverage	Interest Cover Debt Service Coverage ratio
Leverage	Debt equity ratio Total Operating Liabilities to Tangible Net Worth (TOL/TNW)
Liquidity	Current ratio Quick ratio
Firm-specific characteristics	
Size	Size Decile
Age	Year of incorporation
Industry	Industry Group of the firm

techniques required fulfilling the assumptions about the data concerning normality and linearity. The researchers were required to define the structure of the model a priori and then estimate its parameters to fit in the data (Gupta, 2022).

Machine learning techniques appeared in the late 1980s as an alternative and an improvement over the statistical models (Bellovary et al., 2007). Machine Learning techniques are data-driven and tend to be more computationally based. Numerous papers have been published using various machine-learning techniques for distress prediction. They are non-parametric, and despite requiring more computational power, they have become the primary bankruptcy prediction method, outperforming statistical models (Clement, 2020). Machine learning models are good at classifying almost any variable selection method and are more robust to multicollinearity (Liang et al., 2015).

Since both statistical and Machine learning techniques have their own merits and limitations, the study employs one statistical technique and two machine learning techniques to identify the accuracy and performance of each method.

The study applies random forest, artificial neural network, and logistic regression models, which are used to predict the firm's outcome (in terms of bankruptcy or reorganisation)[1] With a certain level of accuracy.

The following techniques were used for data analysis:

- Random Forest (Machine Learning model)—This model is a supervised algorithm that creates a random forest. Many decision trees are combined to make a random forest. It is used in regression and classification problems. This structure is easier to understand, and its interpretation is straightforward. It has certain advantages: the tree structure is visualised, and there is no need for extensive data preparation for analysis (Aker & Karavardar, 2023). Also, the diversity is created by incorporating the benefit of double randomisation, where each tree is estimated based on a synthetic sample drawn randomly from an estimation sample (Klaus-Peter Hellwig, 2021)
- Artificial Neural network (Machine Learning model)—This model adopts the human brain system and can generalise knowledge for predicting future events. This model is used widely for optimisation support and prediction with a back-propagation algorithm (Alamsyah et al., 2021).
- Logistic Regression (Statistical technique)–This model emerged as the most suitable for handling situations wherein the dependent variable is classified or categorical. It helps in determining cause-effect relationships with certain explanatory variables. The prediction in this model is made using equations. This model is focused on finding an equation that would minimise the difference between the dependent variable's actual value and the independent variable's predicted value (Özdamar, 2002). Cross-validation and stratified sampling techniques were adapted to conduct the modelling as specified by Japkowicz and Shah (2011).

4.1 Data Pre-processing

Machine learning models follow the Garbage in, Garbage out (GIGO) principle (Arora & Saurabh, 2022). Therefore, the data must be preprocessed before the modelling stage for accurate results. Preprocessing required dealing with an imbalanced dataset, as the liquidated firms were dominant in number, and spreading the sample to enable random selection of instances. SMOTE, spread subsample, and randomise filter were used to deal with this.

Ten sets were randomly created using 10-fold cross-validation, with 100 iterations performed to get the best results. Cross-validation is a technique for evaluating model performance on unseen data.

[1] Bankruptcy and Liquidation have been used interchangeably.
Restructuring and reorganisation have been used interchangeably as reorganisation is financial restructuring.

4.2 Optimisation Parameters for Model

The study is based on machine learning models that must be tuned for optimum results. Optimizing the hyperparameters to achieve the best possible model architecture is often necessary to implement machine learning algorithms successfully (Beniwal et al., 2023). The hyperparameter optimisation for the machine learning models is as follows-

1. Random Forest—The default batch size was 100, the number of iterations remained the same, and the seed's initial value remained 1.
2. Neural Network—The neural network's learning rate was 0.3, and the batch size was 100. The hidden layers were set to half the sum of attributes and class.

4.3 Performance Evaluation Metrics

The study applied the performance evaluation metrics to evaluate the model's predictive and classification ability: the Confusion Matrix and its indicators, ROC (Receiver Operating Characteristics), and the Gini coefficient.

4.3.1 Prediction Performance (Confusion Matrix) and its Indicators

A confusion matrix gives valuable information about the classifier's performance by comparing actual and predicted classes (Balasubramanian et al., 2019).

A sample confusion matrix is provided in Table 4.

4.3.1.1 Sample Confusion Matrix The reorganised cases have been referred to as positive, and bankrupt cases have been referred to as unfavourable.

Classification accuracy rates calculated from the confusion Matrix as follows-

1. *Accuracy ratio (AR)*: It is the proportion of cases that have been correctly classified (positive or negative).

$$\text{Accuracy Ratio} = (\text{Npp} + \text{Nnn})/\text{Total Sample}$$

   where, Npp(Nnn) is the number of positive(negative) cases classified as positive(negative)
2. *Sensitivity*: It is the proportion of negative cases correctly classified as harmful.

Table 4 Sample Confusion Matrix

Predicted			
Actual		Positive	Negative
	Positive	TP	FN
	Negative	FP	TN

$$\text{Sensitivity} = \text{Nnn}/(\text{Nnn} + \text{Nnp})$$

where Nnn (Nnp) is the number of negative (negative) cases classified as negative(positive)

3. *Specificity*: It is the proportion of positive cases correctly classified as positive and is measured as follows-

$$\text{Specificity} = \text{Npp}/(\text{Npp} + \text{Npn})$$

where, Npp(Npn) is the number of positive(positive) cases classified as positive(negative)

4. *Precision*: is the proportion of correctly classified negative cases out of all those classified as harmful. It is measured as.

$$\text{Precision} = \text{Nnn}/(\text{Nnn} + \text{Npn})$$

where, Nnn(Npn) id number of negative(positive) cases classified as negative(negative)

5. *Type I and Type II errors*: Type I error is the proportion of negative cases classified as positive. In contrast, type II error is the proportion of Positive cases classified as unfavourable.

$$\text{Type I error} = 1 - \text{Sensitivity}$$
$$\text{Type II error} = 1 - \text{Specificity}$$

6. *Gini Coefficient*: The Gini coefficient is an essential tool for assessing the consistency in the results of a machine learning model. It is also the widely used index for validating credit scoring models. (Gupta, 2022).
   It is calculated as G=2(AUROC)–1

Classification rates calculated from the confusion Matrix provide information about assessing a classification model's accuracy and appropriateness. Out of all the parameters discussed above, the most important one from a decision-maker's perspective is the type I error. The reason is that the cost of classifying a negative case as positive far exceeds that of a positive case as unfavourable. For example, the losses to a bank from lending to a bankrupt firm (Type I error) are expected to be much higher than the losses from refusing to lend to a typical going concern, which are the foregone interest payments (Type II error) (Almaskati et al., 2021). It is worth mentioning, however, that the costs to society and the economy from Type II errors might be significant when suitable opportunities are systematically declined due to this error (Almaskati et al., 2021). Thus, error cost should be an essential criterion for model evaluation (Adnan Aziz & Dar, 2006).

4.3.2 Classification Power (Receiver Operating Characteristic (ROC) Curve) and Gini Coefficient

Apart from the confusion matrix and its accuracy rates for prediction ability, the ROC curve or Area under the curve (AUC) and the Gini coefficient are used as classification ability indicators.

The analysis uses the sensitivity (the percentage of default positive outcomes that have been correctly classified) v/s 1- specificity (the percentage of default adverse outcomes that have been correctly classified) to make the curve. The false positive rate is plotted on the horizontal axis, with the actual positive rate on the vertical axis. With this, we get the area under the ROC curve (AUROC) based on a particular model. The area ranges from 0.5 (for a worthless model) to 1 (for a perfect classifier). The larger the area under the curve, the better the model's classification ability.

The Gini Coefficient highlights the difference between a base classifier (taken as the default 0.5 for each class) and the classifier model in question.

5 Results and Discussion

The results indicate the output of a Machine Learning or statistical technique in the form of Accuracy Rate, Precision, Type I error and AUC. The output was generated by first including only financial variables and later incorporating firm-specific characteristics with the economic variables.

The analysis was spread over five years before the date of Liquidation/restructuring. Also, the study was carried out each year by including financial variables only and later on by including financial variables and firm-specific characteristics. It was observed that the accuracy rate was higher when firm-specific characteristics were included along with the economic variables to predict the financial distress outcome of the firm. Also, the accuracy declines as we move further away from the date of liquidation/restructuring. However, accuracy cannot be considered as the criteria for measuring the prediction accuracy of a model. According to (Hsieh, 1993). The cost linked with Type I error is higher for the stakeholders than the cost related to Type II error, so the best model is the one with the lowest Type I error.

The results of the various models one year before bankruptcy/reorganisation are given in Table 3. As we compare Panels A and B, we observe that the accuracy rate, precision, area under the curve (AUC), and Gini coefficient provide a better classification and prediction measure when the firm-specific characteristics are included with the financial variables. Out of the three models employed in the study, the Random Forest is the best bet in the distress prediction models as it has the lowest Type-I error. As per Anandarajan et al., (2001), "The objective of learning classifications from sample data is to classify and predict successfully on new data. A classifier's error rate is the most commonly used measure of success or failure. The error rate is statistically defined as the error rate of the classifier on an asymptotically large number of new cases that converge in the limit to the actual population distribution." For the analysis results, one year before the actual event of bankruptcy/restructuring, as shown in Table 5. It can be seen that type-I error is

Table 5 One year before Bankruptcy/reorganisation

Model	Accuracy rate	Precision	Type I error (1-sensitivity)	AUC	Gini coefficient
Panel A: Only financial variables					
Random Forest	**0.613**	0.586	**0.087**	**0.631**	**0.262**
Neural network	0.557	0.559	0.233	0.565	0.13
Logistic	0.543	0.543	0.146	0.518	0.036
Panel B: Financial variables and firm-specific characteristics					
Random Forest	**0.735**	0.711	**0.160**	**0.784**	**0.568**
Neural network	0.727	0.736	0.244	0.740	0.48
Logistic	0.718	0.708	0.204	0.764	0.528

The bold text signifies the best out of all the values signifying the superiority of a model over the others based on that specific parameter

lower in the analysis done when considering only financial variables. One reason for that could be that as the firm approaches bankruptcy or restructuring, the financial position of the firm, as exhibited in the financial statements, has already deteriorated to a great extent. An analysis of attributes using the attribute selector indicates that size, Return on Total Assets, and Interest cover are the significant attributes impacting the firm's failure. Thus, as the firm nears failure, the ability to pay the financial creditors becomes a major deciding factor in terms of the fate of the firm.

As we move further away from the date of liquidation and restructuring, we see (Table 6) that the accuracy rate of the model declines. Also, the probability of type-I error increases. This highlights that distress prediction is more accurate one year before liquidation/restructuring. Also, the attributes impacting the final decision concerning the firm's fate are size, and Gross fixed asset utilisation ratio. The size of the firm plays a significant role, and this follows the results as defined by Malakauskas and Lakštutienė, (2021) and Arora and Saurabh, (2022).

Table 6 Two years before Bankruptcy/Reorganisation

Model	Accuracy rate	Precision	Type I error (1-sensitivity)	AUC	Gini coefficient
Panel A: Only financial variables					
Random Forest	**0.577**	0.568	**0.164**	**0.617**	**0.234**
Neural network	0.557	0.581	0.414	0.604	0.208
Logistic	0.522	0.531	0.166	0.546	0.092
Panel B: Financial variables and firm-specific characteristics					
Random Forest	**0.736**	0.715	**0.166**	**0.789**	**0.578**
Neural network	0.723	0.729	0.241	0.745	0.490
Logistic	0.723	0.711	0.196	0.754	0.508

The bold text signifies the best out of all the values signifying the superiority of a model over the others based on that specific parameter

(1994) opine that a firm is in financial trouble if its Earnings before interest, tax, depreciation, and amortisation (EBITDA) are less than its expenses. Therefore, the current study examines three years before the firm is adjudicated to be liquidated or restructured and calculates the accuracy rate and other prediction and classification indicators.

As we look at the results of Table 7, we see better prediction accuracy and a lower type-I error. This signifies that the prediction based on financial variables and firm-specific characteristics was more accurate three years before the firm was declared bankrupt or eligible for restructuring.

Thus, it can be analysed that to predict a firm's financial distress, it is imperative to look into its performance as soon as it starts showing warning signals regarding low income and inability to pay operational creditors. Three years before the judgement of the distressed firms as bankrupt or restructured, the significant attributes/variables that are important are size and Profit after tax as a percentage of total income.

Early detection of a firm's movement towards failure in the early stages benefits business stakeholders. (Ashraf et al., 2019) In all the years of analysis, random forest emerged as a clear winner in predicting early distress and had the lowest Type I error, indicating its superiority over the traditional Logit model. This can be due to the fundamental difference between traditional statistical and Machine Learning models in handling data and making predictions. Statistical models, such as linear and logistic regression, are based on predefined assumptions about data distribution and variable relationships. These models are typically parametric, assuming a specific functional form (e.g., linearity) to define the relationship between input variables and the outcome. Statistical models are often effective with smaller datasets and perform well when the relationships among variables are well-understood. However, their predictive accuracy can be limited if data distribution assumptions are unmet. Here, the dataset comprises 1111 firms, which is a large dataset in terms of statistical models. On the other hand, Machine-learning models focus on flexible learning from complex, high-dimensional data, resulting in enhanced predictive capabilities. ANN

Table 7 Three years before Bankruptcy/Reorganisation

Model	Accuracy rate	Precision	Type I error (1-sensitivity)	AUC	Gini coefficient
Panel A: Only financial variables					
Random Forest	**0.607**	0.590	**0.153**	**0.665**	0.33
Neural network	0.591	**0.627**	0.437	0.653	**0.306**
Logistic	0.546	0.554	0.270	0.567	0.134
Panel B: Financial variables and firm-specific characteristics					
Random Forest	**0.729**	**0.709**	**0.174**	**0.782**	**0.564**
Neural network	0.701	0.707	0.259	0.739	0.478
Logistic	0.707	0.698	0.215	0.745	0.490

The bold text signifies the best out of all the values signifying the superiority of a model over the others based on that specific parameter

follows Random Forest, indicating the superiority of machine learning models over conventional prediction models. The study's results are similar to those reported by Gupta, (2022), Balasubramanian et al., (2019), and Sehgal et al., (2021).

6 Summary of Findings

The analysis indicates that predicting a firm's financial distress necessitates early detection through evaluating performance indicators, particularly when warning signs such as declining income and an inability to meet obligations to operational creditors become apparent. This would enable the firm to move towards sustainable measures in terms of innovation and creativity so as to deal with the ever-growing and dynamic business environment (Jain & Gupta, 2018; Sinha et al., 2009). The predictive accuracy of distress models improves considerably when firm-specific characteristics, including age, industry, and size, are incorporated alongside financial variables. Among the models assessed, Random Forest demonstrated the highest predictive efficacy, followed by neural networks and logistic regression, underscoring the superior performance of machine learning algorithms compared to traditional statistical models in forecasting financial distress. Firm size emerged as a particularly influential determinant, with smaller firms exhibiting a greater vulnerability to liquidation, consistent with prior findings by Almaskati et al., (2021) and Sehgal et al., (2021). Furthermore, the analysis revealed a decline in predictive accuracy as the forecasting horizon extended, aligning with observations by Charalambakis and Garrett (2019) and Sehgal et al., (2021). Key financial variables influencing distress prediction were profit after tax as a percentage of total income, return on assets and interest coverage. Notably, the cost associated with Type I errors exceeded that of Type II errors, emphasising the importance of models with minimised Type I errors. Finally, while statistical models offer interpretability through predefined relationships, machine learning models leverage flexibility in learning from complex, high-dimensional data, enhancing predictive capabilities.

7 Conclusion

Financial distress leads to significant economic losses for all stakeholders, making it imperative to identify early distress signals. It encompasses several adverse consequences, including reduced market share and reputational standing, increased capital costs, a withdrawal of favourable trade credit terms from suppliers, and losses associated with asset sales conducted under distressed conditions (Li, 2024). Additionally, financial distress often leads to the departure of valuable human capital and heightened exposure to aggressive strategies from competitors. This paper aimed to determine the predictive and classification ability of the traditional logit model with the machine learning models. It also aimed at determining the relative importance of the various attributes in predicting distress. Classic financial ratios proposed by Turetsky and McEwen, (2007), Altman et al., (2017) and Almaskati et al., (2021) were used, along

with the firm's specific characteristics, such as its age, size, and industry. A comparative analysis was done to identify the impact of including non-financial variables in the model.

The traditional logit model was compared to two machine learning models: Random forest and Artificial Neural network. The predictive ability of the three models was compared using the AUC under the ROC curve and the type-I error. The Random Forest model emerged as the clear winner, followed by ANN and the logit model, thereby highlighting the supremacy of machine learning models over traditional models. The analysis also identified the highest predictive ability years before the firm failed. It was found that three years before the liquidation/restructuring of the firm, the Random Forest model had the highest predictive accuracy and lowest error rate. It is also observed that all the models had a predictive accuracy greater than 75%, thus confirming their robustness.

The study involved the firms that were referred to NCLT, and to the best of our knowledge, it is the first paper using the database of only distressed firms as compared to an equivalent number of solvent firms in line with the study undertaken by Balasubramanian et al., (2019).

To predict financial distress, it is imperative to look into the firm's size and significant financial variables like Profit after tax as a % of total income, Return on assets and Interest coverage. This can aid the managers in assessing the economic health of their firms and serve as a significant indicator for the investors who are willing to invest in such firms. It can significantly contribute to the banking sector, helping them identify the default firms and save their balance sheets from the Non-Performing Asset (NPA) problem.

Further, the analysis can be extended by incorporating macroeconomic factors such as inflation, GDP growth, etc. Machine learning algorithms like Naïve Bayes can be used to perform sentiment analysis of other components of Annual reports like Management discussion and analysis (MD&A). A Naïve Bayes classifier assigns a probability to the word or text, whether positive or negative (Arora & Saurabh, 2022). A detailed industry-based study can also be undertaken to predict specific patterns of financial distress in the Indian economy. This study included all companies restructured or liquidated by the NCLT, regardless of their industry. An industry-specific analysis, however, could provide more comprehensive insights and yield more robust results. The study is based on the financial ratios derived from CMIE Prowess. Also, the missing ratios, which are irrelevant and can't be computed for a particular industry, were imputed by the machine-learning software WEKA based on the median values. Such adversities can be handled when the study concerns a particular industry.

Declarations

Conflict of interest The authors declare no conflict of interest.

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